In [ ]:

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import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import mean_squared_error, accuracy_score

# Load and prepare the data
df =
pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

# For simplicity, we'll drop rows with missing age
df.dropna(subset=['Age'], inplace=True)

# Define features and target
features = ['Age', 'Pclass']
target_reg = 'Fare'
X_reg = df[features]
y_reg = df[target_reg]

# Split the data
X_train_reg, X_test_reg, y_train_reg, y_test_reg = train_test_split(X_reg,
y_reg, test_size=0.2, random_state=42)

# --- ENTER YOUR CODE HERE ---
# 1. Create the model instance
lr_model = LinearRegression()

# 2. Train the model
lr_model.fit(X_train_reg, y_train_reg)

# 3. Make predictions
y_pred_reg = lr_model.predict(X_test_reg)

# 4. Evaluate the model
mse = mean_squared_error(y_test_reg, y_pred_reg)
print(f"Mean Squared Error for Fare Prediction: {mse:.2f}")
print(f"The square root of this is {np.sqrt(mse):.2f}, meaning our model is
off by about $48 on average.")


# Prepare data for classification
# We need to encode 'Sex' column
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})

# Define features and target
features_cls = ['Age', 'Pclass', 'Sex']
target_cls = 'Survived'
X_cls = df[features_cls]
y_cls = df[target_cls]

# Split the data
```

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X_train_cls, X_test_cls, y_train_cls, y_test_cls = train_test_split(X_cls,
y_cls, test_size=0.2, random_state=42)

# --- ENTER YOUR CODE HERE ---
# 1. Create the model instance
log_model = LogisticRegression()

# 2. Train the model
log_model.fit(X_train_cls, y_train_cls)

# 3. Make predictions
y_pred_cls = log_model.predict(X_test_cls)

# 4. Evaluate the model
accuracy = accuracy_score(y_test_cls, y_pred_cls)
print(f"Accuracy for Survival Prediction: {accuracy:.2%}")
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Mean Squared Error for Fare Prediction: 3364.92
The square root of this is 58.01, meaning our model is off by about $48 on
average.
Accuracy for Survival Prediction: 74.83%
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Knowledge Check

1. In your own words, what is the key difference between a regression problem and a classification problem?

Regression is for predicting actual numbers (like a price, temperature, or fare), while classification is for sorting things into categories or classes (like whether someone survived or not, yes/no, or spam/not spam).

2. The LinearRegression model has an attribute called .coef_. After you train the model, print lr.model.coef. What do these numbers represent?

Those numbers are the coefficients (or weights) for each of our features (like Age and Pclass). They basically tell us how much each feature impacts the predicted fare—showing the strength and direction of the relationship between that feature and the target.

3. Why did we use mean_squared_error to evaluate the regression model but accuracy score the classification model? Why wouldn't accuracy be a good metric for the fare prediction task?

We use MSE for regression because we're measuring how far off our predicted numbers are from the actual values (and squaring the errors penalizes bigger mistakes). Accuracy only works when you're checking if a discrete category prediction is right or wrong. For fare prediction, getting the exact dollar amount right down to the penny is super rare, so if we used accuracy, it would pretty much always be 0%. MSE actually tells us how close our guesses were on average.